PROJECT TASKFLOW PRO

EPISODE:B03

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CHAPTER 1: Problem Statement

- Task Overload: incorrect Prioritization
- Data Isolation
- Data Serialization: Lightweight DB

  → NoSQL Document

CHAPTER 2: Objectives

- To-Do List: Category, Priority, Priority
- JSON Serialization → RESTful API → MongoDB
- Authentication & User Isolation: JWT and Bcrypt
- External API: Third-party Quotes API

CHAPTER 3: Data Serialization Theory

- Serialization: In-Memory Object → JSON String
- Deserialization: JSON String → In-Memory Object
- Why JSON: Lightweight, Fast, Human-Readable Standard
- Native BSON Mapping with MongoDB Document Store

CHAPTER 4: TECHNOLOGY STACK

MERN STACK ARCHITECTURE

- MongoDB Atlas: Cloud NoSQL Database (Singapore Region)
- Express.js: RESTful API Backend Framework
- React + Vite: Glassmorphism SPA (Dark/Light Mode)
- Node.js Runtime & Render Cloud Platform Deployment
- IDE: Antigravity - AI Gemini 3.8 Flash

CHAPTER 5: JSON MAP STRUCTURE

- User Schema: _id, name, email, password (Bcrypt Hash)
- Todo Schema: _id, user (Ref), title, category, priority, completed
- User Isolation: Enforced by user ObjectId Reference
- Standard JSON Format: Key-Value, Numbers, Booleans, Arrays

CHAPTER 6: API MAP STRUCTURE

- POST /api/auth/register & /api/auth/login (Auth)
- GET /api/todos (Read & Real-time Search)
- POST / PUT / DELETE /api/todos/:id (Full CRUD)
- GET /api/quotes/random (External API Proxy)

CHAPTER 7: USER SCENARIOS & SCOPE

- Guest User: Landing Page, Quote Inspiration, Register/Login
- Authenticated User: Personal Dashboard & Real-time Stats
- Task Operations: Create, Filter, Search, Edit, Toggle, Delete
- UI Customization: Instant Dark Mode & Light Mode

CHAPTER 8: LIVE DEMONSTRATION

- Live URL: https://taskflow-pro-aeq6.onrender.com
- User 1: student1@ku.th / Password123!
- User 2: student2@ku.th / Password123!
- Demo Sequence: CRUD Flow → Search & Filter → User Isolation Test

CHAPTER 9: CONCLUSION & Q&A

- Complete MERN Stack Application with 100% Functional CRUD
- Seamless JSON Data Serialization & Cloud Deployment
- Clean Code Architecture with Thai Comments
- Thank You — Q&A Session

THE END OF PRESENTATION

FINALE:

Thank you for your attention.